



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ROYAL AGENCIES SALEM

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

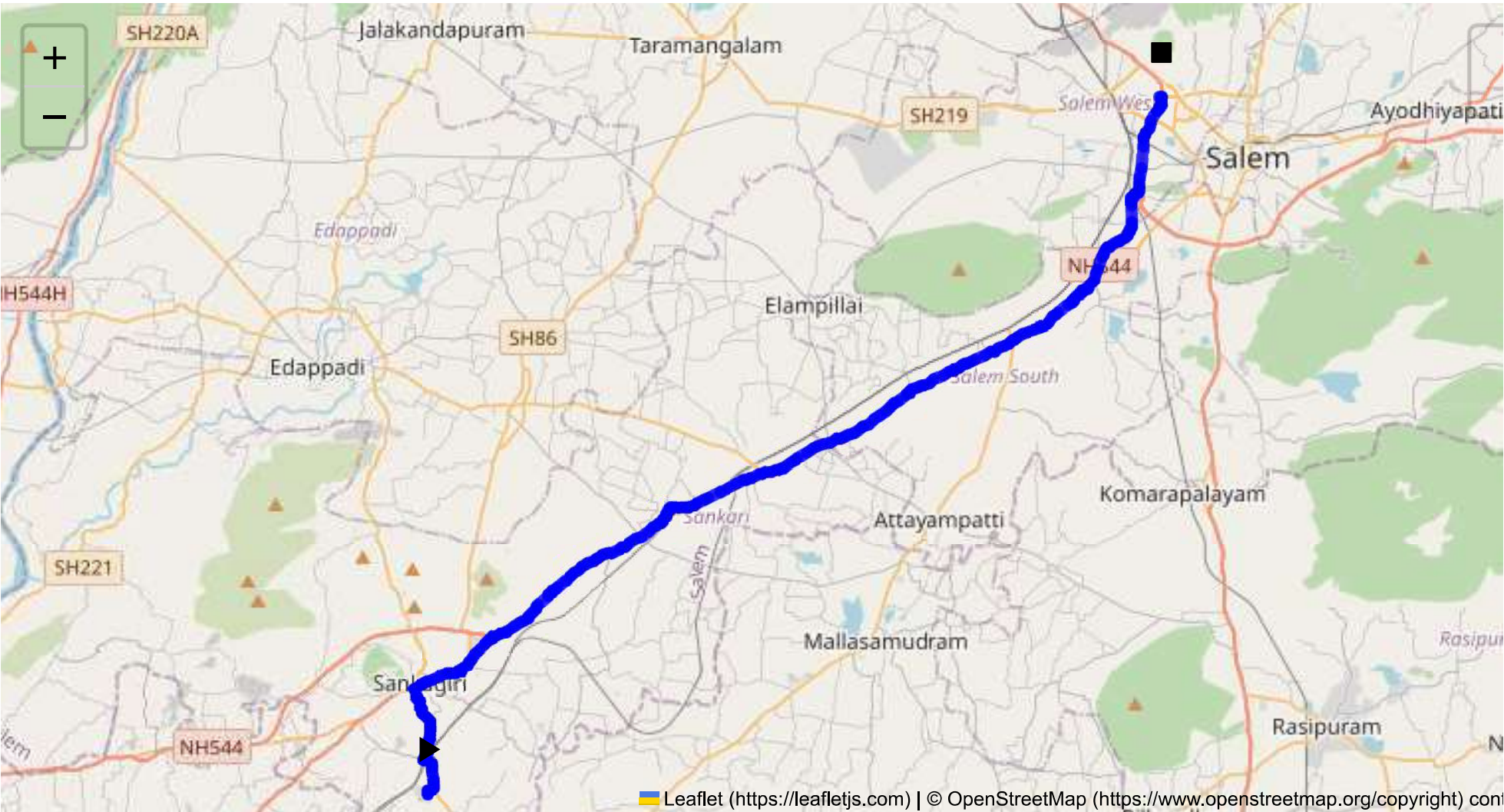
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.06 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.675472, 78.127133)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 645, Junction Main Rd, Salem, is approximately 43.06 kilometers long. The journey primarily follows the NH 544 (previously known as NH 47), which is a major highway

connecting these locations. This highway is utilized frequently for commercial transport, including heavy vehicles.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. The typical weather patterns include:

- **Summer (March to June):** High temperatures up to 40°C, potential for road melt and heat-related vehicle issues.
- **Monsoon (July to September):** Heavy rains leading to potential flooding and reduced visibility.
- **Post-Monsoon (October to November):** Occasional rains, can make roads slippery.
- **Winter (December to February):** Mild temperatures, generally safer conditions but occasional fog in early mornings may reduce visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8-10 AM) and Evening (5-7 PM) coinciding with industrial shifts and commuting.
- **Congestion-prone Areas:** Entry and exit points near Salem and intersections at Toll Plazas are hotspot areas for congestion.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** NH 544 is generally well-maintained, with proper signage and regular maintenance.
- **Infrastructure:** Adequate bridges and flyovers but be vigilant of occasional potholes and uneven surfaces, particularly after monsoons.

5. Suggestions for Alternative Routes for Emergencies

- **Via SH 86:** Divert through State Highway 86 to bypass major congestion.
- **Via Local Roads:** Utilize Uthamasolapuram Church Road for localized diversions avoiding major highway stretches during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Timing Restrictions:** Certain roads might have restrictions during peak hours for hazardous transport.
- **Documentation:** Ensure all necessary permissions are acquired from local authorities regarding hazardous material transport.

7. Overview of Historical Incidents

- **Heavy Vehicle Incidents:** There have been records of rollovers and minor collisions, typically due to overspeeding and abrupt braking.

- **Hazardous Material Incidents:** Limited reporting but potential risks are commonly associated with improper containment and securement of materials.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** No major environmentally sensitive zones on the direct route, but caution advised near agricultural areas to prevent spillage.
- **Water Bodies:** Bridges over small watercourses require care to avoid contamination risk in case of a spill.

9. Analysis of Communication Coverage

- **Coverage:** Generally good cellular and network coverage along NH 544, with limited dead zones.
- **Dead Zones:** Potential weak signal areas in interior roads and valleys near hilly regions.

10. Estimated Emergency Response Times

- **Highway Segments:** Approx. 20-30 minutes for emergency services response due to good highway access.
- **Interior Roads:** Response times can extend to 40-60 minutes in more rural or congested areas.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Salem via NH 544 is relatively safe but requires due diligence concerning weather conditions and traffic patterns. Maintaining proper vehicle checks and adhering to speed guidelines can mitigate many risks. Alternative routes should be pre-planned to avoid delays in emergencies, and drivers must be well-informed of local regulations affecting hazardous materials transport. Overall, risk is moderate, but proactive measures can ensure safe transit.

It is recommended for drivers to communicate regularly with dispatch regarding route conditions and to utilize GPS systems for real-time traffic updates. Regular training on emergency procedures and incident management will further enhance safety outcomes.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48366, 77.88781	30 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.60942, 78.10056	30 KM/Hr
10	Turn	Medium	11.61736, 78.10588	30 KM/Hr
11	Turn	Medium	11.61753, 78.10608	30 KM/Hr
12	Turn	Medium	11.64007, 78.11794	30 KM/Hr
13	Turn	Medium	11.64020, 78.11786	30 KM/Hr
14	Turn	Medium	11.64497, 78.12028	30 KM/Hr
15	Turn	Medium	11.64503, 78.12034	30 KM/Hr
16	Turn	High	11.67074, 78.12668	15 KM/Hr
17	Turn	High	11.67081, 78.12661	15 KM/Hr
18	Blind Spot	Blind Spot	11.67570, 78.12786	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
25	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
27	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
28	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
26	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47407, 77.86839



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48366, 77.88781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60942, 78.10056



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64007, 78.11794



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64020, 78.11786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64497, 78.12028



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64503, 78.12034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67074, 78.12668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67081, 78.12661



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67570, 78.12786

